

$\Rightarrow$ the product xy equals to zero iff $x=0$ or $y=0$.

$$\frac{xy=0}{P} \Leftrightarrow \frac{(x=0)}{Q} \vee \frac{(y=0)}{R}$$

$$\rightarrow P \Leftrightarrow Q \vee R \leftarrow$$

P	Q	R
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

$Q \vee R$
T
T
T
F
T
T
T
F

$P \Leftrightarrow (Q \vee R)$
T
T
T
F
F
F
F
T

P	Q
T	T
T	F
F	T
F	F

$$xy = 0$$

$$x = 0$$

$$y = 0$$

$$xy = 6 \quad \begin{matrix} x = 2 \\ y = 3 \end{matrix}$$

$$P \Leftrightarrow Q \vee R$$

$$\downarrow \quad \downarrow$$

$$A \quad B$$